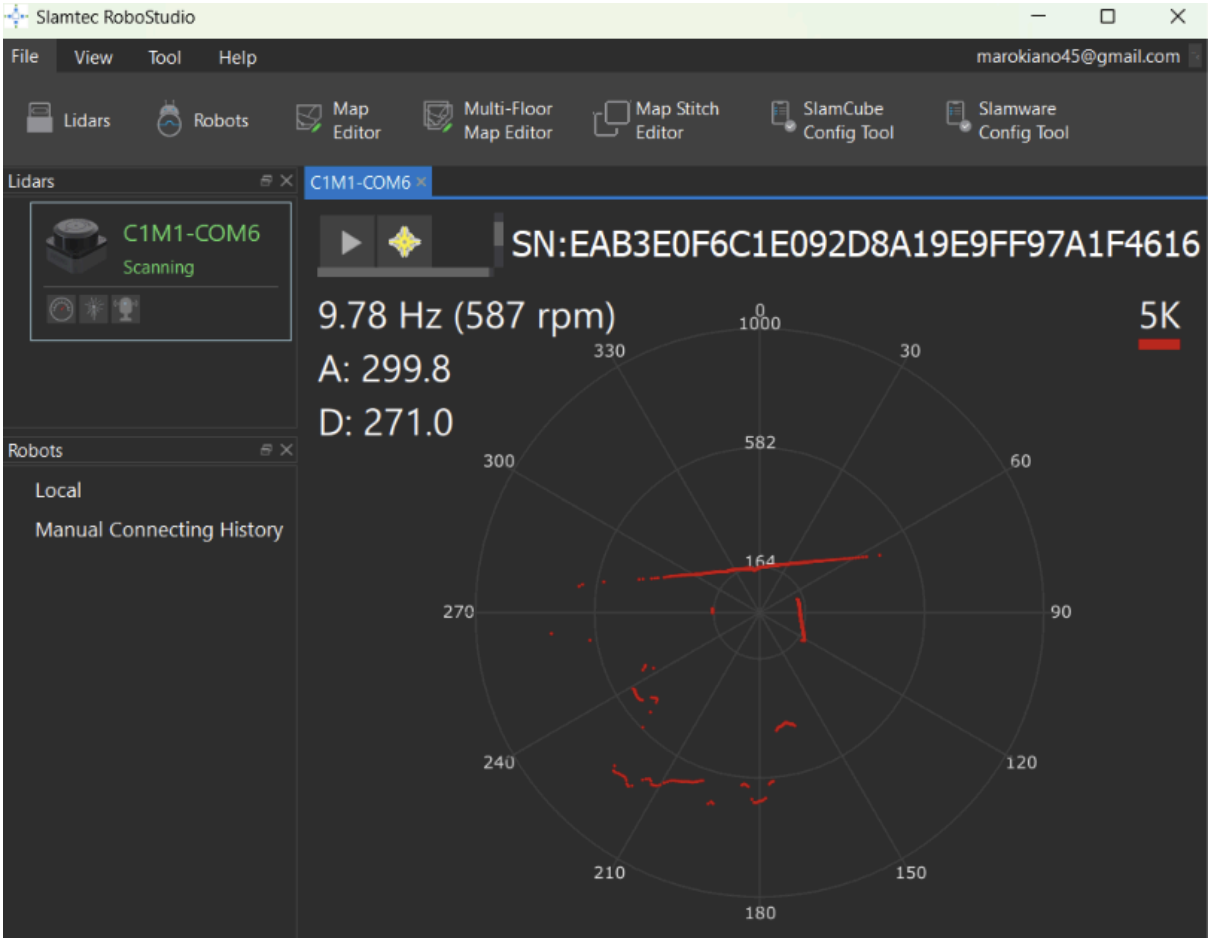
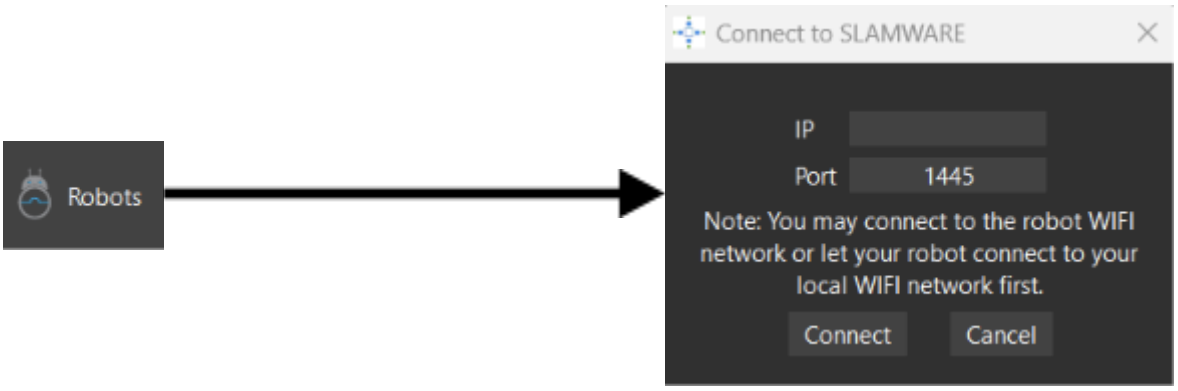
ref : dfr1138
dfrobot
rplidar c1 360° laser range scanner 12 M

Power supply: 5 V DC
- In operation: 230 mA
Interface: TTL-UART (USB converter included): Rx, Tx, GND
Front (marked "RPLIDAR"): 0-degree position
Average frequency: 9.8 Hz (587 rpm)

1st test using the Slamtec app from RoboStudio





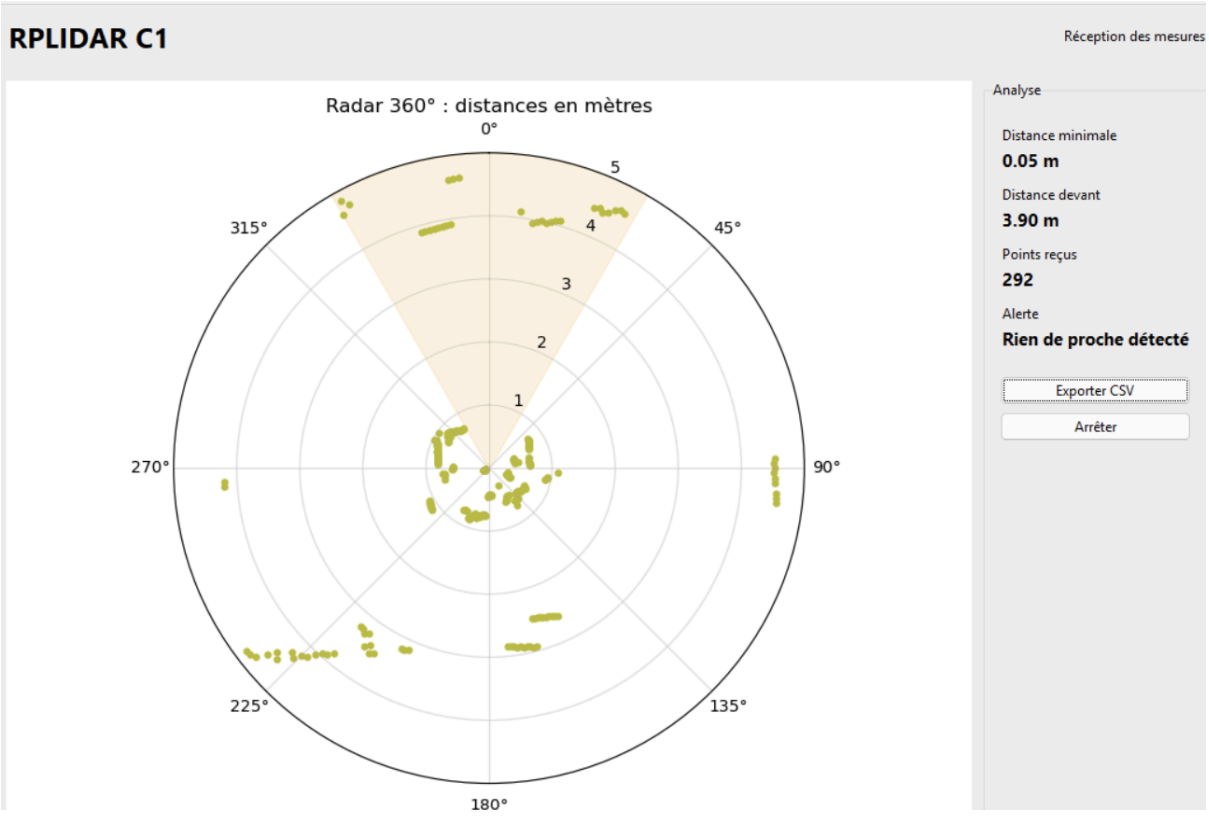
https://github.com/dsaadatmandi/rplidarc1

Inertial Measurement Unit (IMU): Records the precise orientation of the sensor as it moves, following the robot's construction.

Frequency: 10 Hz

Based on a .py module

Initial attempt at visualization from VS Code using Python



Web interface + Python

Accessible from anywhere—especially useful for outdoor use, such as via a smartphone.
Justification for choosing a web interface for the GyroBot

Opting for a web interface—developed using HTML, CSS, and JavaScript, and powered by a Python server running on a Raspberry Pi 5—allows for the separation of supervisory functions from the robot's control and data-processing tasks. This architecture enables access to the interface from various devices over a local network, eliminating the need to install specific software on each individual device.

Furthermore, it allows for the reuse of existing Python programs—such as those for LiDAR data acquisition and map processing—while providing an interactive visualization of the map and the robot's status. Decoupling the graphical interface, SLAM processing, and motor control simplifies both system maintenance and future development.

Web interface, Python-based mapping, JavaScript, Flask server.

New features will likely be developed through collaboration with the rest of the team; initial impressions of the various site pages are already underway.

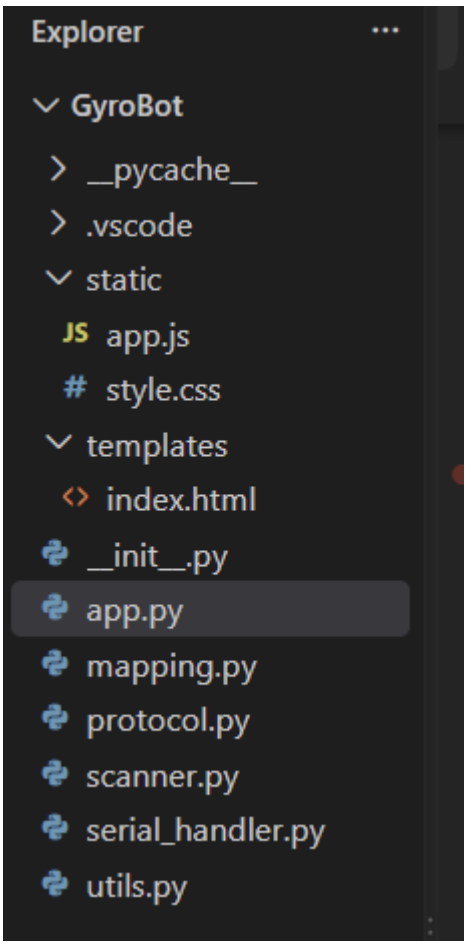
Flask: A Python framework for building web servers.
It handles the creation and configuration of your web server.

In the GyroBot project, `app.js` makes the web interface interactive and dynamic.
It is a JavaScript file that executes directly within the user's web browser (e.g., Chrome, Firefox).
It acts as a bridge between user actions, the HTML display, and the data provided by Flask.

The RPLIDAR C1 LiDAR sensor collects distance and angle measurements from the robot's surroundings. Python programs (specifically `scanner.py` and `mapping.py` from the GitHub repository) retrieve and process this data to calculate the coordinates of obstacles.

Flask (`app.py`) then serves as an intermediary between the Python backend and the web interface. It transmits the data to the JavaScript layer, which renders it as a graphical representation within the browser. The user can thus view LiDAR measurements, and start or stop the scan from their computer or phone, without needing to interact directly with the Python code.

In summary: LiDAR → Python → Flask → Web interface.



`mapping`, `protocol`, `scanner`, `serial_handler`, and `utils` are modules from a GitHub repository that provides a Python library (a Python driver for the RPLIDAR C1).
Its purpose is to provide the functions necessary to communicate with the LiDAR, sparing the user from having to manage low-level data exchanges directly.

`scanner.py`: Communicates with the LiDAR and retrieves measurements.
`mapping.py`: Uses these measurements to calculate the positions of obstacles.
`app.py` (Flask): Transmits the data to the web interface.
`app.js`: Displays the data on the web page.

Not Secure http://172.28.54.29:5000

GyroBot

Robot autonome et cartographie

• Serveur connecté

Accueil | Cartographie | Contrôle | Données

Tableau de bord

Bienvenue sur l'interface du GyroBot.

Prêt à explorer ? Prenez les commandes du GyroBot et laissez-le vous faire découvrir son environnement.

État du robot

Supervision

LiDAR

En attente

Distance minimale

-- m

Batterie

-- %

GyroBot | Interface de supervision

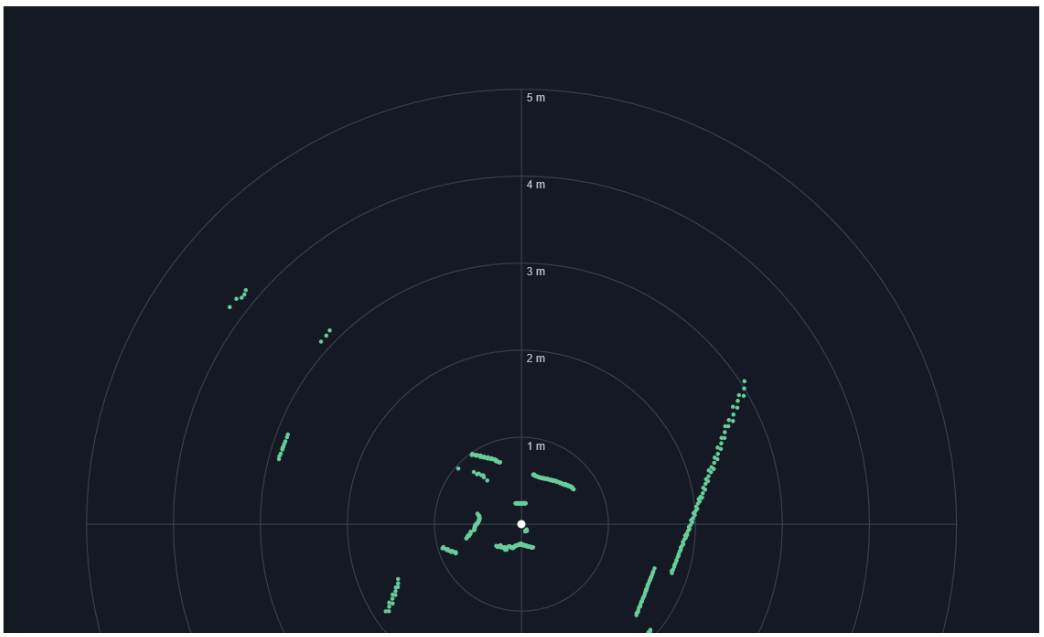
Robot autonome et cartographie

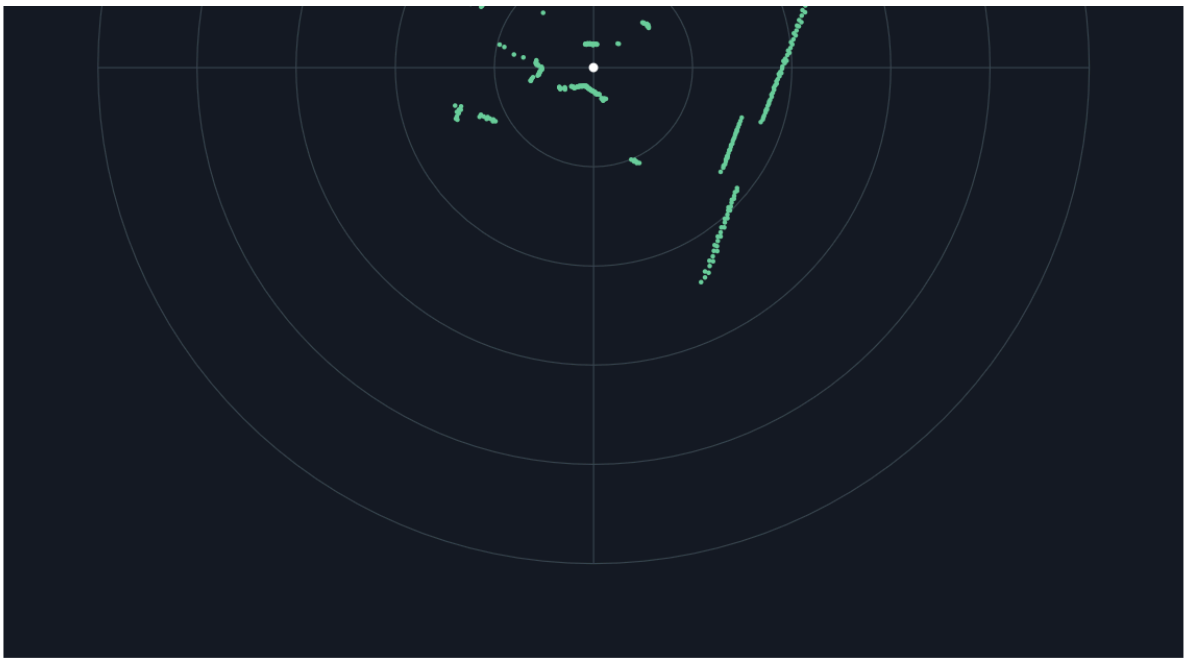
• Serveur connecté

Accueil | Cartographie | Contrôle | Données

Cartographie LiDAR

Carte de l'environnement





Scan LiDAR en cours

Démarrer le mapping | Arrêter le mapping

Mesures du LiDAR

Distance devant : 1.79 m

Nombre de mesures : 445

GyroBot

Robot autonome et cartographie

• Serveur connecté

Accueil | Cartographie | Contrôle | Données

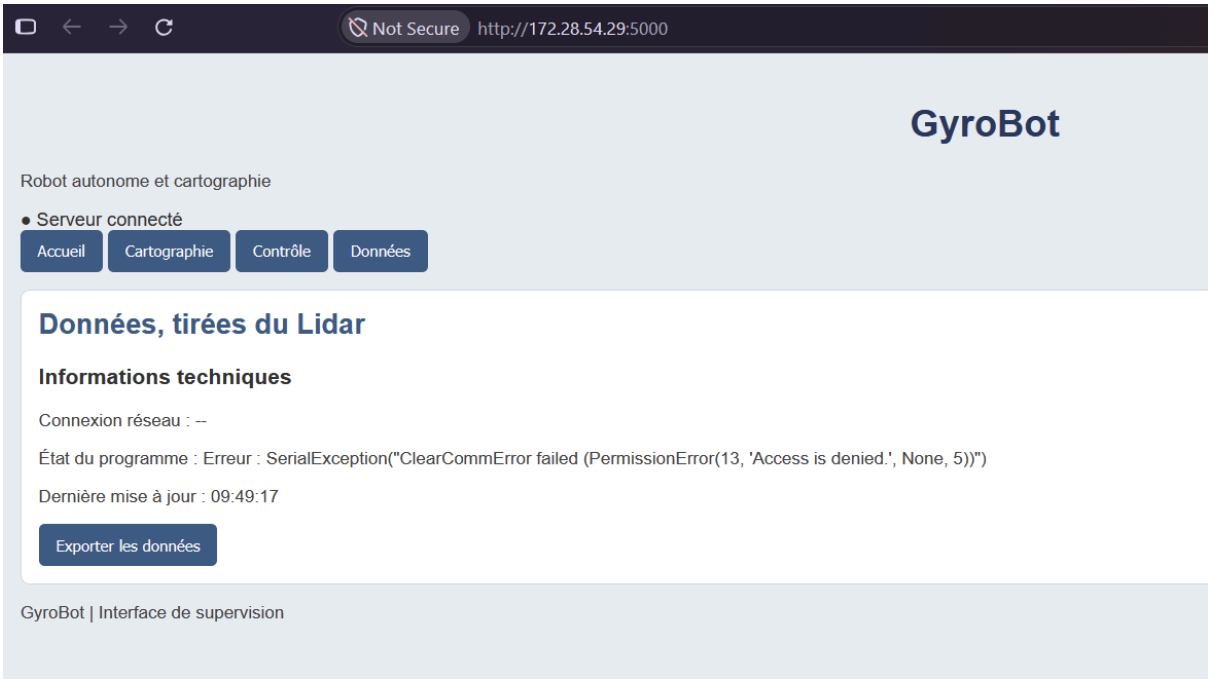
Contrôle du robot

Déplacement manuel

▲
◀ ■ ▶
▼

Les commandes seront activées après la connexion au système de contrôle du robot.

GyroBot | Interface de supervision


Not Secure
http://172.28.54.29:5000
GyroBot
Robot autonome et cartographie
• Serveur connecté
Accueil
Cartographie
Contrôle
Données
Données, tirées du Lidar
Informations techniques
Connexion réseau : --
État du programme : Erreur : SerialException("ClearCommError failed (PermissionError(13, 'Access is denied.', None, 5))")
Dernière mise à jour : 09:49:17
Exporter les données
GyroBot | Interface de supervision